to see, but the UP action also has a spike on top of the NO-OP spike on the left, so the UP action-value distribution is bimodal, suggesting that taking the UP action might lead to either a –1 reward or a +10 reward, but the +10 reward is more likely since that spike is taller.

Figure 7.22. Each column has the action-value distributions for a particular action for a given state (row). The number above each plot is the expectation value for that distribution, which is the weighted average value for that distribution. The distributions look fairly similar by eye, but the expected values are distinct enough to result in significantly different action selections.

Figure 7.23. In quantile regression, rather than learning what probabilities are assigned to a fixed set of supports, we learn a set of supports that correspond to a fixed set of probabilities (quantiles). Here you can see that the median value is 1 since it is at the 50th percentile.

Chapter 8. Curiosity-driven exploration

Figure 8.1. Screenshot from the Montezuma's Revenge Atari game. The player must navigate through obstacles to get a key before any rewards are received.

Figure 8.2. In environments with dense rewards, the rewards are received fairly frequently during the training time, making it easy to reinforce actions. In sparse reward environments, rewards may only be received after many sub-goals are completed, making it difficult or impossible for an agent to learn based on reward signals alone.

Figure 8.3. Prediction error is summed with the extrinsic environment reward for use by the agent.

Figure 8.4. The prediction module takes in a state, St, (and action at, not shown) and produces a prediction for the subsequent state, Št+1 (pronounced "S hat t+1", where the hat symbol suggests an approximation). This prediction, along with the true next state, are passed to a mean-squared error function (or some other error function), which produces the prediction error.

Figure 8.5. The noisy TV problem is a theoretical and practical problem where a reinforcement learning agent with a naive sense of curiosity